

Module 14: Testing (Unit, Feature, Browser)

Duration: 7-10 days | **Level:** Intermediate-Advanced

PHPUnit Setup

Laravel includes PHPUnit. Run tests:

```
php artisan test
php artisan test --filter=TaskTest
php artisan test --coverage
```

Feature Tests

Test HTTP endpoints and full flows:

```
class TaskTest extends TestCase
{
    use RefreshDatabase;

    public function test_user_can_create_task(): void
    {
        $user = User::factory()->create();
        $project = Project::factory()->create(['user_id' => $user->id]);

        $this->actingAs($user)
            ->post(route('tasks.store'), [
                'project_id' => $project->id,
                'title' => 'New task',
            ])
            ->assertRedirect();

        $this->assertDatabaseHas('tasks', ['title' => 'New task']);
    }
}
```

API Tests

```
Sanctum::actingAs($user);

$this->getJson('/api/tasks')
    ->assertOk()
    ->assertJsonCount(3, 'data');
```

Unit Tests

Test isolated classes without HTTP:

```
class DashboardServiceTest extends TestCase
{
    use RefreshDatabase;

    public function test_stats_returns_correct_counts(): void
    {
        $service = app(DashboardService::class);
        // ...
    }
}
```

Factories in Tests

```
$user = User::factory()->create();
Task::factory()->count(5)->overdue()->create([
    'user_id' => $user->id,
]);
```

Mocking

```
Mail::fake();
Notification::fake();
Queue::fake();
Event::fake();

// Assert
Mail::assertSent(WelcomeMail::class);
Queue::assertPushed(SendTaskDueReminder::class);
```

Browser Tests (Laravel Dusk)

```
composer require laravel/dusk --dev
php artisan dusk:install

$this->browse(function (Browser $browser) {
    $browser->visit('/login')
    ->type('email', 'admin@taskforge.test')
    ->type('password', 'password')
    ->press('Sign In')
    ->assertPathIs('/dashboard');
});
```

TaskForge Test Suite

File	Tests
tests/Feature/AuthTest.php	Login, register, logout
tests/Feature/TaskTest.php	CRUD, complete, scopes
tests/Feature/Api/TaskApiTest.php	Sanctum API

Testing Best Practices

1. Use RefreshDatabase for isolation
 2. One assertion concept per test
 3. Descriptive test names: test_guest_cannot_view_tasks
 4. Test happy path AND edge cases
 5. Run tests in CI on every push
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Exercises

1. Add test: member cannot delete admin's project.
2. Add test: completing task fires TaskCompleted event.

3. Achieve 80%+ coverage on controllers.

Next Module

-> [Module 15: Security](#)